

Jordan Crisis Management Simulation Engine

9XAI Fellows Scope Package

Simulation-First National Crisis Intelligence and Decision Support System

1. Mission

Build a fully functional simulation-first national crisis management platform capable of operating today on synthetic crisis data sources and later transitioning to live operational data sources without redesigning the core platform.

This is not a dashboard exercise. This is not a reporting platform.

The system must behave like a national crisis intelligence and decision support platform that receives signals, normalizes data, detects emerging risks, identifies root causes, simulates cascading impacts, recommends interventions, evaluates outcomes, and supports decision-makers during complex national events.

The platform should help leaders answer:

- What is happening?
- Why is it happening?
- What will happen next?
- What should we do?
- What happens if we choose a different response?

2. Macro Architecture

The system must contain four layers.

Layer 1: Crisis Source Layer

- Synthetic crisis feeds
- Simulated agency APIs
- Injected crisis events
- Future production integrations
- Advanced external signal providers

All sources must be treated through the same onboarding, validation, mapping, and ingestion logic.

Layer 2: Integration and Control Layer

- Source registry
- API discovery

- Schema validation
- Source health monitoring
- Source activation
- Source decommissioning
- Simulation lifecycle management

Layer 3: Crisis Intelligence Core

- Canonical crisis model
- Risk engine
- Anomaly detection
- Dependency engine
- Root cause engine
- Simulation engine
- Prediction engine
- Intervention engine
- Impact estimation engine

Layer 4: Experience Layer

- National Crisis Cockpit
- Decision Hub
- Simulation Console
- Signal Explorer
- Source Management Console
- Conversational Intelligence Interface

3. Crisis Domains to Simulate

Teams may select one or more domains.

Water Security

- Reservoir depletion
- Pipeline failures
- Pumping station outages
- Distribution disruption

Public Health

- Disease outbreaks
- Hospital overload
- Medicine shortages
- Emergency response saturation

Infrastructure

- Power outages
- Telecom failures
- Transportation disruptions
- Utility failures

Disaster Response

- Flash floods
- Earthquakes
- Wildfires
- Severe weather

Supply Chain

- Fuel shortages
- Food shortages
- Border disruptions
- Logistics failures

Social Stability

- Misinformation campaigns
- Panic behavior
- Service access challenges
- Trust deterioration

4. Synthetic Data Requirements

The sandbox environment must generate realistic crisis-related data. Examples include: emergency incidents, service outages, weather alerts, resource shortages, citizen distress signals, agency response logs, infrastructure failures, traffic disruptions, mobility restrictions, and intervention records.

Data should contain:

- Missing fields
- Delayed reporting
- Duplicate events
- Conflicting signals
- Cascading impacts
- Misleading symptoms
- False positives
- Seasonal variation
- Geographic clustering
- Hidden dependencies

The data must be difficult enough to test intelligence capabilities.

5. Sandbox API Layer

Every sandbox source must expose APIs. Each API should define:

- Endpoint and request/response structure
- Required and optional fields
- Timestamps and source identifier
- Location identifier and event identifier
- Severity level and pagination
- Error handling and freshness timestamp

The sandbox APIs must be discovered, validated, mapped, and consumed exactly as future production systems.

6. Dynamic Source Onboarding

The platform must allow future crisis sources to be added without rebuilding the core platform. New sources must support:

- Registration
- Metadata discovery
- Schema validation
- Field mapping
- Data quality assessment
- Activation
- Intelligence propagation
- Dashboard propagation

Minimum demonstration: Add at least one source not originally part of the sandbox and demonstrate it affects risk calculations, simulations, recommendations, and operational insights.

7. Canonical Crisis Data Model

The system must normalize all sources into a unified model. Core objects include:

- Crisis event, hazard, location, agency
- Asset, resource, infrastructure component
- Population segment, incident, alert, signal
- Dependency, risk, simulation, intervention
- Outcome, recommendation, prediction, root cause, decision

Teams may extend the model.

8. National Risk Index

The platform must create a National Risk Index. The index may combine:

- Incident severity
- Response readiness
- Infrastructure stress
- Resource availability
- Public sentiment
- Service disruption
- Recovery capacity
- Population exposure

The index should be calculated at national, governorate, agency, service, and crisis levels, and explain what changed, why it changed, contributing factors, dominant risks, and recommended actions.

9. Intelligence Core

The intelligence loop is: Signal → Insight → Root Cause → Simulation → Recommendation → Action → Outcome

Classification

Classify incident type, severity, urgency, affected sector, geographic area, and criticality.

Pattern Detection

Detect incident spikes, geographic clusters, resource shortages, service deterioration, anomalous behavior, and dependency failures.

Root Cause Analysis

For every major insight generate: likely root cause, supporting evidence, conflicting evidence, confidence score, and missing information warnings.

Prediction

Forecast incident growth, resource depletion, service overload, infrastructure stress, and recovery timelines.

Recommendation Engine

Generate actions with expected impact, urgency, implementation complexity, responsible agency, dependencies, and confidence score.

10. Decision Hub

The Decision Hub converts intelligence into action. Each action should include:

- Triggering event

- Affected risk indicator and systems
- Recommended intervention
- Responsible agency and due date
- Escalation logic
- Expected and actual outcome

11. Simulation Engine

The simulation engine is a separate architectural block. It must support:

- Event injection
- Cascading impact modeling
- Hidden root causes
- Competing interventions
- Before-and-after comparisons
- Simulation rollback
- Scenario expiration

Each simulation should include: trigger, affected entities, affected assets, affected population, propagation logic, misleading signals, hidden cause, expected impact, intervention options, and recovery logic.

12. Minimum Simulations

The platform must demonstrate at least three simulations.

Demand Overload

A sudden surge overwhelms public services.

Infrastructure Failure

A critical asset fails and causes cascading impacts.

Cross-Agency Dependency Failure

One agency failure creates secondary failures in other agencies.

13. Wicked Problem Simulation

A wicked problem simulation is mandatory. The simulation must demonstrate:

- Symptoms appearing in one location
- Root causes originating elsewhere
- Multiple agencies affected
- Changing citizen behavior

- Conflicting KPIs
- Misleading interpretations

Example

A pumping station failure reduces water availability. Hospitals experience higher demand. Call centers experience increased complaints. Traffic congestion increases around water distribution points. The visible symptom appears to be healthcare overload. The actual root cause is water infrastructure failure. The system should identify the true cause and recommend appropriate intervention.

14. Advanced Signal Integration

The platform should support external signals generated outside the core platform. Examples include:

- Crowd density signals
- Mobility signals
- Weather intelligence
- Social sentiment signals
- Infrastructure monitoring signals
- Emergency responder telemetry
- IoT sensor networks

The system does not generate these signals. The system consumes, validates, and integrates them.

15. Creativity Zone

Teams are encouraged to innovate beyond the minimum scope. Examples:

- Digital Twins
- AI Crisis Advisors
- Agent-Based Simulation
- Resource Optimization Engines
- Autonomous Response Planning
- National Wargaming Systems
- Multi-Agent Decision Intelligence
- Predictive Crisis Forecasting
- Adaptive Intervention Planning

Innovation beyond the stated requirements is encouraged and will be rewarded.

16. Minimum Acceptance Criteria

The final system must include:

- At least five simulated source types
- At least three crisis domains
- Working sandbox APIs
- Source onboarding
- Canonical crisis model
- Risk engine
- Simulation engine
- Root cause analysis
- Recommendation engine
- National Crisis Cockpit
- Decision Hub
- Event injection
- Three simulations
- One wicked problem simulation
- One advanced signal source
- Dynamic source onboarding
- Simulation cleanup
- Production transition strategy

17. High-Bar Outcome

A strong build will deliver:

- Reusable national crisis digital twin
- Simulation control console
- Explainable risk intelligence
- Advanced signal integration
- Root cause confidence scoring
- Intervention impact estimation
- Decision intelligence cockpit
- Simulation rollback capability
- Multi-agency coordination framework
- Executive-grade crisis command platform

18. Final Output

The final output should prove that:

- Crisis intelligence systems can be built before all integrations are available
- Synthetic sources can emulate operational systems
- Simulations can stress-test national readiness
- Advanced signals improve crisis awareness

- Interventions can be evaluated before deployment
 - Decision-makers can explore multiple response strategies
 - Simulation data can be isolated and removed safely
 - The platform can evolve continuously into a national crisis management capability
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High-Bar Vision

A future Jordan Crisis Management Platform that enables leaders to understand, anticipate, simulate, and manage complex national crises before consequences escalate.